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Department of Computer Engineering

Course Name:	Web Development Laboratory (316U01L306)	Semester:	III
Date of Performance:	8/9/2026	DIV/ Batch No:	A2
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Experiment No: 4

Title

Form Validation using JavaScript for a Climate Intelligence and Heatwave Early Warning Web Portal

Aim of the Experiment

To design and develop a JavaScript-based form validation module for a Climate Intelligence for Heatwave Monitoring, Prediction, and Early Warning web portal, ensuring that user-entered weather monitoring, stakeholder, and alert-related information is complete, valid, and correctly formatted.

Objectives of the Experiment

- To create an interactive HTML form for collecting heatwave monitoring and alert-related information.
- To validate form fields using JavaScript event handling and conditional statements.
- To apply Regular Expressions for validating name, mobile number, email address, and AWS station ID.
- To display meaningful error messages for invalid inputs and confirmation messages for valid inputs.
- To demonstrate client-side validation as part of a climate intelligence decision-support platform.

COs to be achieved

CO1: Developing webpages using HTML, CSS and JavaScript.

Books / Journals / Websites references

1. MDN Web Docs – JavaScript and HTML Form Validation

2. W3schools.com

Theory

Form validation is the process of checking whether data entered by a user satisfies predefined rules before it is submitted. JavaScript provides client-side validation using events, conditional statements, functions, string

methods, and Regular Expressions. It gives immediate feedback and helps prevent incomplete or incorrectly formatted data.

In this experiment, validation is applied to a Climate Intelligence and Heatwave Early Warning web portal. The use case combines meteorological observations, IoT-enabled Automated Weather Stations (AWS), AI-driven heatwave analytics, visualization, and advisory mechanisms. The form collects station, location, temperature, humidity, date, observer, and alert information.

Problem Statement

"Climate Intelligence for Heatwave Monitoring, Prediction, and Early Warning"

Develop a JavaScript-enabled web form for collecting localized weather monitoring information and stakeholder details for a heatwave monitoring and early warning platform. The form should validate mandatory fields and ensure that temperature, humidity, email, mobile number, AWS station ID, and observation date are entered in the correct format.

Task 1: Create the Heatwave Monitoring Form

Scenario

Design a web form for entering localized heatwave monitoring information.

- Observer Name, Email, Mobile Number, AWS Station ID, Location, Observation Date, Maximum Temperature, Humidity, and Heatwave Alert Level.
- Use suitable HTML5 input types such as text, email, tel, date and number.
- Use labels, fieldset and legend.
- Mark essential fields as required.

Concepts Tested

- HTML Forms, HTML5 Input Types, Form Controls, Labels and Fieldset

Deliverable: heatwave-form.html

Task 2: Validate User Name and Station ID

Scenario

Ensure that the observer name and AWS Station ID contain valid characters and are not blank.

Requirements

- Name must contain alphabets and spaces only and have at least 3 characters.
- AWS Station ID should follow a format such as AWS001.
- Display an error message for invalid values.

Concepts Tested

- JavaScript Functions, Conditional Statements, Regular Expressions, String Validation

Task 3: Validate Email and Mobile Number

Scenario

Validate contact information before accepting the monitoring form.

Requirements

- Validate email using a Regular Expression.
- Validate a 10-digit Indian mobile number.
- Display field-level error messages.

Concepts Tested

- Regular Expressions, JavaScript String Methods, Error Messages

Task 4: Validate Weather Observation Data

Scenario

Ensure temperature and humidity observations are within reasonable input ranges.

Requirements

- Maximum Temperature must be numeric.
- Humidity must be between 0 and 100 percent.
- Observation date must not be empty.
- Display messages for missing or out-of-range values.

Concepts Tested

- Number Validation, Range Checking, Date Validation, JavaScript Conditions

Task 5: Validate Alert Level and Submit Form

Scenario

Validate the Inputs and display a success message when all inputs are correct.

Requirements

- Display a success message only when all rules are satisfied.

Concepts Tested

- Form Events, submit Event, preventDefault(), DOM Manipulation, Client-side Validation

Deliverable: heatwave-validation.html

Code: (Javascript only)

```
const form = document.getElementById("heatwaveForm");

function showError(fieldId, message) {
    document.getElementById(fieldId).textContent = message;
}


function clearErrors() {
    document.getElementById("nameError").textContent = "";
    document.getElementById("emailError").textContent = "";
    document.getElementById("mobileError").textContent = "";
    document.getElementById("stationError").textContent = "";
    document.getElementById("locationError").textContent = "";
    document.getElementById("dateError").textContent = "";
    document.getElementById("temperatureError").textContent = "";
    document.getElementById("humidityError").textContent = "";
    document.getElementById("alertError").textContent = "";
    document.getElementById("successMessage").textContent = "";
}


function validateName() {
    const name = document.getElementById("observerName").value.trim();
    const namePattern = /^[A-Za-z ]+$/;


    if (name === "") {
        showError("nameError", "Observer Name is required.");
        return false;
```

```
    }

    if (name.length < 3) {
        showError("nameError", "Observer Name must contain at least 3 characters.");
        return false;
    }

    if (!namePattern.test(name)) {
        showError("nameError", "Observer Name must contain alphabets and spaces only.");
        return false;
    }

    showError("nameError", "");
    return true;
}

function validateStationId() {
    const stationId = document.getElementById("stationId").value.trim();
    const stationPattern = /^AWS[0-9]{3}$/;

    if (stationId === "") {
        showError("stationError", "AWS Station ID is required.");
        return false;
    }

    if (!stationPattern.test(stationId)) {
        showError("stationError", "Station ID must be in the format AWS001.");
```

```
        return false;
    }

    showError("stationError", "");
    return true;
}

function validateEmail() {
    const email = document.getElementById("email").value.trim();
    const emailPattern = /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/;

    if (email === "") {
        showError("emailError", "Email Address is required.");
        return false;
    }

    if (!emailPattern.test(email)) {
        showError("emailError", "Please enter a valid Email Address.");
        return false;
    }

    showError("emailError", "");
    return true;
}

function validateMobile() {
    const mobile = document.getElementById("mobile").value.trim();
```

```
    const mobilePattern = /^[0-9]{10}$/;

    if (mobile === "") {
        showError("mobileError", "Mobile Number is required.");
        return false;
    }

    if (!mobilePattern.test(mobile)) {
        showError("mobileError", "Mobile Number must contain exactly 10 digits.");
        return false;
    }

    showError("mobileError", "");
    return true;
}

function validateLocation() {
    const location = document.getElementById("location").value.trim();

    if (location === "") {
        showError("locationError", "Location is required.");
        return false;
    }

    showError("locationError", "");
    return true;
}
```

```
function validateDate() {
    const observationDate = document.getElementById("observationDate").value;

    if (observationDate === "") {
        showError("dateError", "Observation Date is required.");
        return false;
    }

    showError("dateError", "");
    return true;
}

function validateTemperature() {
    const temperature = document.getElementById("maxTemperature").value.trim();

    if (temperature === "") {
        showError("temperatureError", "Maximum Temperature is required.");
        return false;
    }

    if (isNaN(temperature)) {
        showError("temperatureError", "Maximum Temperature must be numeric.");
        return false;
    }

    showError("temperatureError", "");
```

```
    return true;
}

function validateHumidity() {
    const humidity = document.getElementById("humidity").value.trim();

    if (humidity === "") {
        showError("humidityError", "Humidity is required.");
        return false;
    }

    if (isNaN(humidity)) {
        showError("humidityError", "Humidity must be numeric.");
        return false;
    }

    const humidityValue = Number(humidity);

    if (humidityValue < 0 || humidityValue > 100) {
        showError("humidityError", "Humidity must be between 0 and 100 percent.");
        return false;
    }

    showError("humidityError", "");
    return true;
}
```

```
function validateAlertLevel() {
    const alertLevel = document.getElementById("alertLevel").value;

    if (alertLevel === "") {
        showError("alertError", "Please select a Heatwave Alert Level.");
        return false;
    }

    showError("alertError", "");
    return true;
}

form.addEventListener("submit", function (event) {
    event.preventDefault();

    clearErrors();

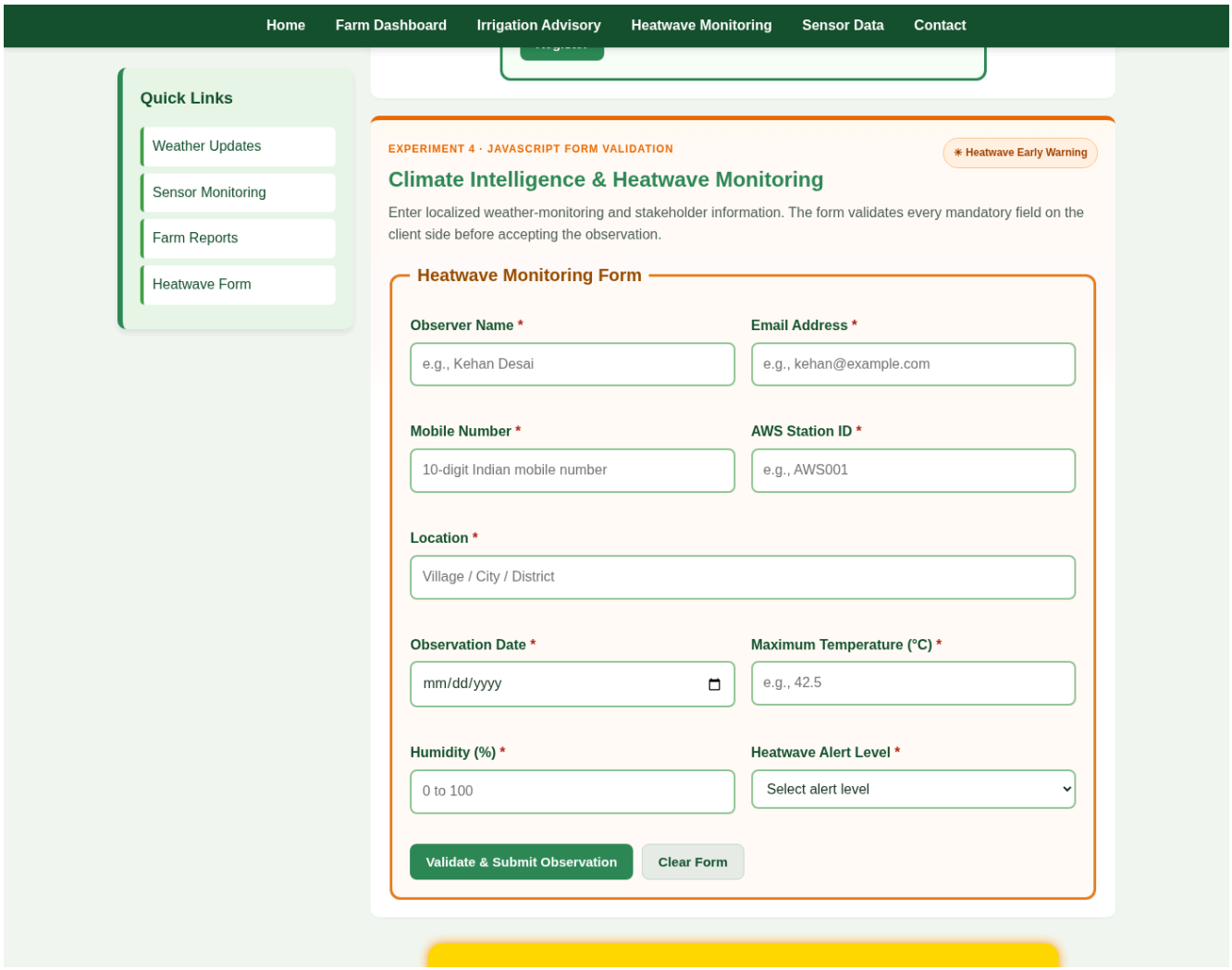
    const isNameValid = validateName();
    const isEmailValid = validateEmail();
    const isMobileValid = validateMobile();
    const isStationValid = validateStationId();
    const isLocationValid = validateLocation();
    const isDateValid = validateDate();
    const isTemperatureValid = validateTemperature();
    const isHumidityValid = validateHumidity();
    const isAlertValid = validateAlertLevel();
```

```
    if (
        isNameValid &&
        isEmailValid &&
        isMobileValid &&
        isStationValid &&
        isLocationValid &&
        isDateValid &&
        isTemperatureValid &&
        isHumidityValid &&
        isAlertValid
    ) {
        document.getElementById("successMessage").textContent =
            "Heatwave Monitoring Information submitted successfully!";

        document.getElementById("successMessage").style.color = "green";
    }
});

document.getElementById("observerName").addEventListener("blur", validateName);
document.getElementById("email").addEventListener("blur", validateEmail);
document.getElementById("mobile").addEventListener("blur", validateMobile);
document.getElementById("stationId").addEventListener("blur", validateStationId);
document.getElementById("location").addEventListener("blur", validateLocation);
document.getElementById("observationDate").addEventListener("change", validateDate);
document.getElementById("maxTemperature").addEventListener("blur", validateTemperature);
document.getElementById("humidity").addEventListener("blur", validateHumidity);
document.getElementById("alertLevel").addEventListener("change", validateAlertLevel);
```

Output / Screenshot



Home | Farm Dashboard | Irrigation Advisory | Heatwave Monitoring | Sensor Data | Contact

Quick Links

- Weather Updates
- Sensor Monitoring
- Farm Reports
- Heatwave Form

EXPERIMENT 4 · JAVASCRIPT FORM VALIDATION

✳ Heatwave Early Warning

Climate Intelligence & Heatwave Monitoring

Enter localized weather-monitoring and stakeholder information. The form validates every mandatory field on the client side before accepting the observation.

Heatwave Monitoring Form

Observer Name *
K1
Use alphabets and spaces only, with at least 3 characters.

Email Address *
abc@
Enter a valid email address.

Mobile Number *
12345
Enter a valid 10-digit Indian mobile number.

AWS Station ID *
ST01
Use AWS followed by 3 digits, for example AWS001.

Location *
Village / City / District
Location is required.

Observation Date *
mm/dd/yyyy
Observation date is required.

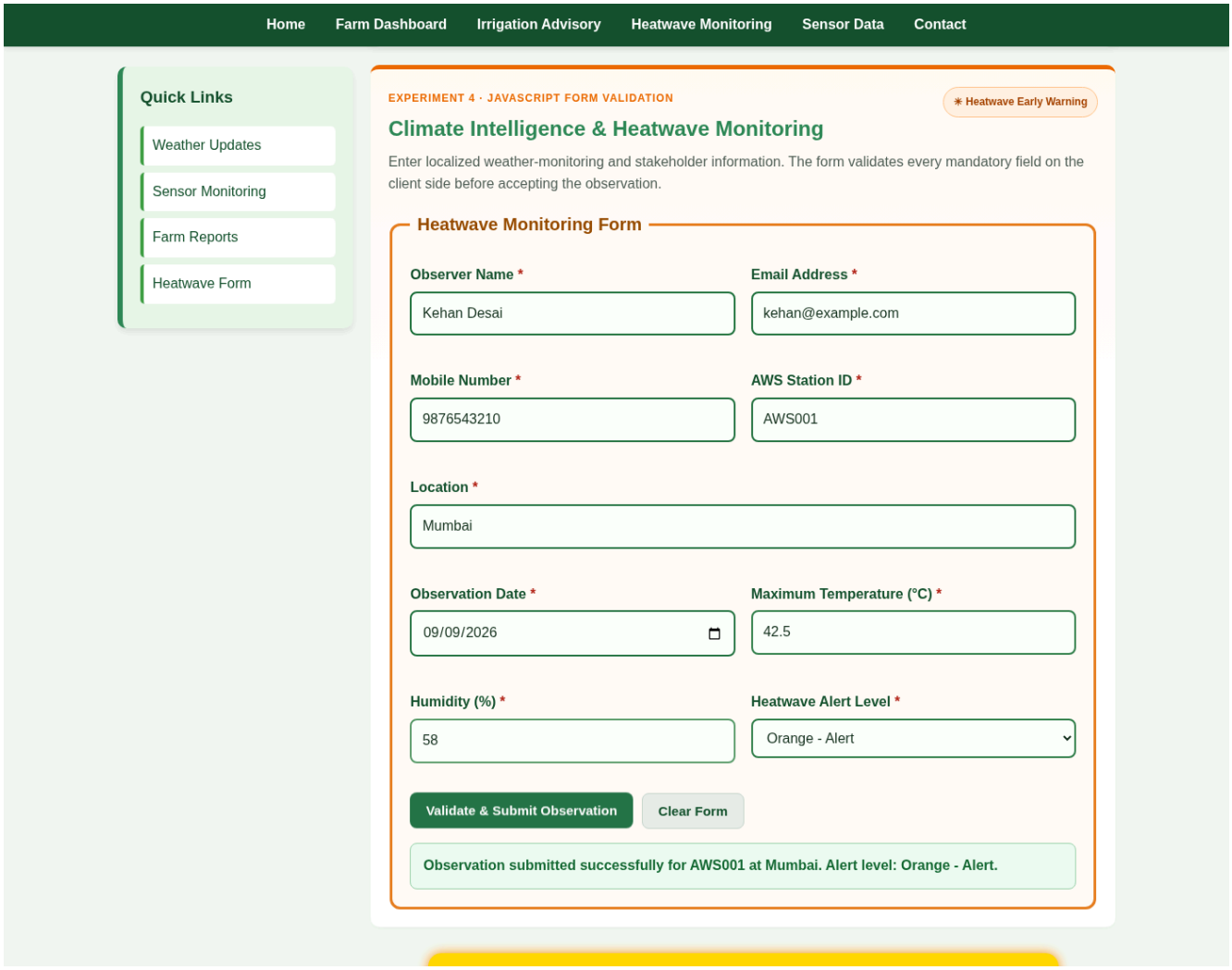
Maximum Temperature (°C) *
e.g., 42.5
Maximum temperature is required.

Humidity (%) *
120
Humidity must be between 0 and 100 percent.

Heatwave Alert Level *
Select alert level
Select a heatwave alert level.

Validate & Submit Observation | Clear Form

Please correct the highlighted fields before submitting the observation.



Students should execute the HTML file in a web browser and capture screenshots showing:

- The completed Heatwave Monitoring form.
- Validation error messages for invalid inputs.
- Successful submission message for valid inputs.z

Conclusion and Discussion

The experiment demonstrates how JavaScript can be used for client-side form validation in a web-based Climate Intelligence system. Validation rules ensure that monitoring and stakeholder information is complete and correctly formatted before processing. The experiment demonstrates JavaScript functions, events, conditional statements, Regular Expressions, DOM manipulation, and preventDefault(). The module can serve as a front-end component of a larger heatwave monitoring, visualization, advisory, and early warning platform.